Website Development Agent-Based for Administration Automation of GJM and GKM Vocational Faculty

PRESENTED BY

GROUP 06 43TRPL

Member Group

11423009	**Odelia Josephine Simanjuntak**
11423032	**Ray Anugerah Tua Simarmata**
11423040	**Daniel Sinambela**
11423042	**Frangky Hamonangan Sirait**
11423059	**Venesa Herawaty Hutajulu**

BACKGROUND

GKM and GJM are responsible for academic quality assurance, but all administrative processes are still done manually using Excel and email.

Key Issues:

- No integrated system
- Manual monitoring & recapitulation
- Manual reminders
- Reports created from scratch
- Does Not have a centralized data management

CURENT SYSTEM

GKM and GJM administrative processes are done manually using Excel and email. GKM officers check RPS and material data from CIS, write completeness using Excel, contact lecturers via email, recapitulate questionnaires manually, and create reports by copying and pasting data from various sources. These reports are then sent to GJM via email.

GKM: Check from CIS → Write Excel → Contact lecturers → Manual questionnair recap → Create report → Email to GJM

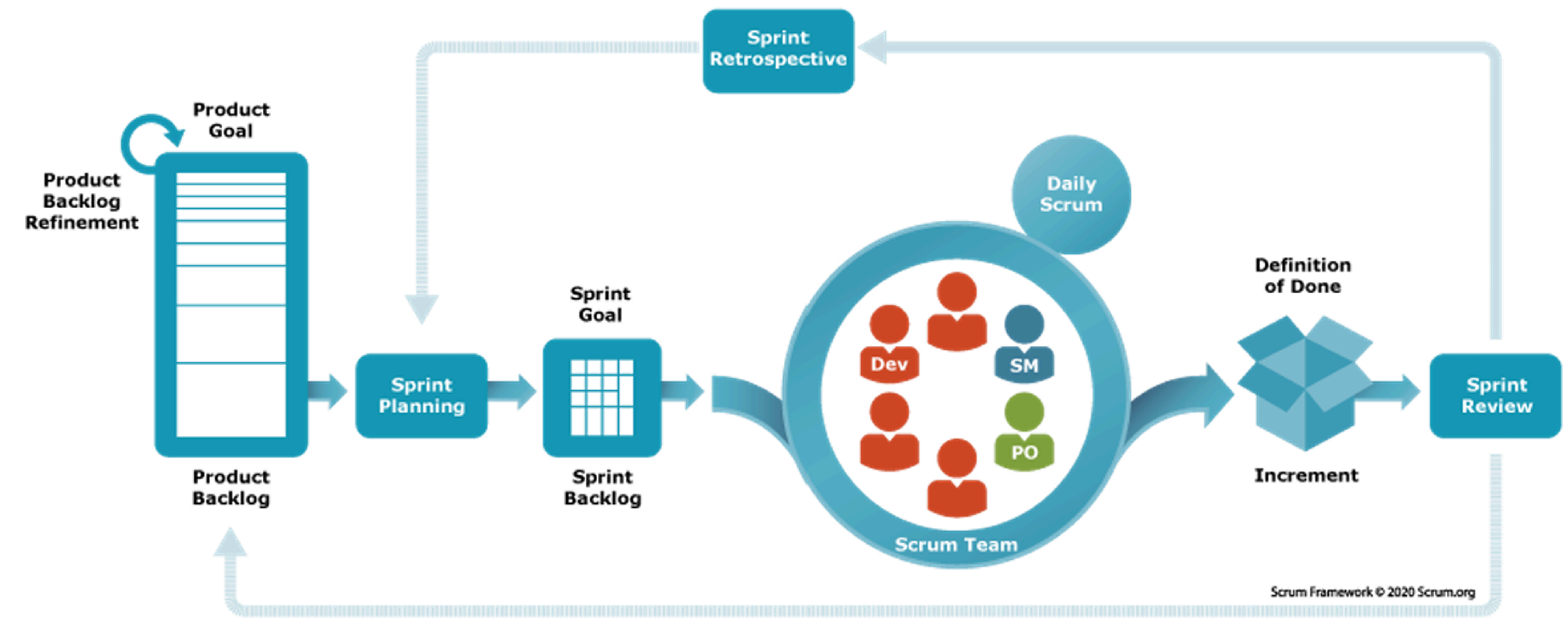
receive emails from multiple GKM units, manually recapitulate the data, create quarterly and semester reports, make PowerPoint presentations manually, and finally send everything to the Faculty Leadership.

GJM: Receive email → Manual recap → Create quarterly reports → Manual PPT → Send to Leadership

TARGET SYSTEM

- To develop an integrated web-based system for managing GKM and GJM administrative data.
- To automate repetitive administrative processes
- To provide accurate, well-documented, and easily accessible information.
- To support digital transformation in quality assurance administration.

METHODOLOGY



- In this project, the Agile Development methodology with the Scrum framework is used to support iterative, flexible, and adaptive system development in response to changing user requirements.

PROJECT PLANNING OVERVIEW

At the beginning of the project, we planned the system development by identifying user needs, defining the product goal, and preparing the product backlog. We also divided the team roles using the Scrum method, selected the development tools, estimated the project scope, and identified possible risks. This planning helped the team develop the system in a structured, organized, and controlled way.

PRODUCT BACKLOG

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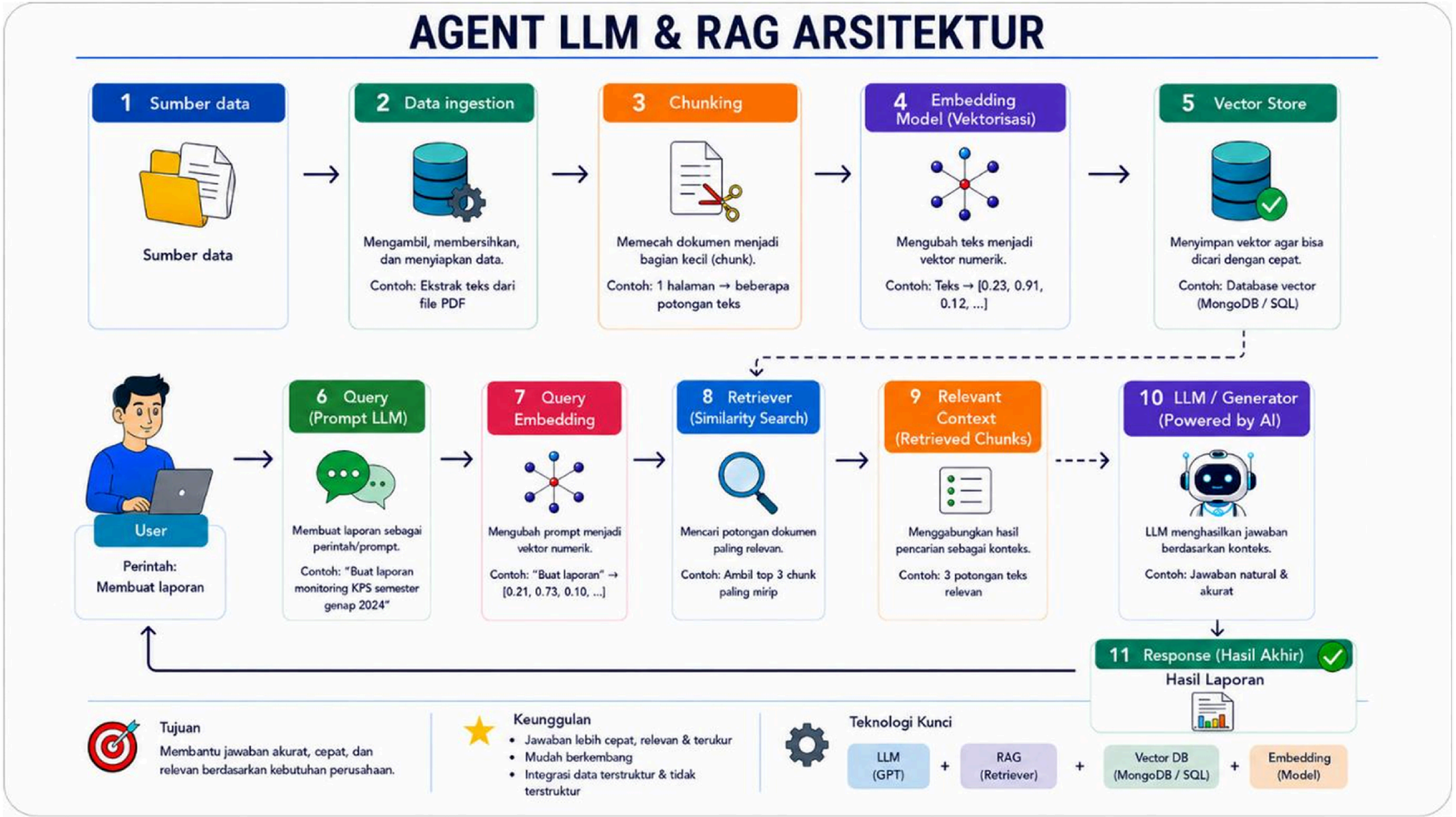
Target Sprint	PBI ID	Nama Backlog	Tanggal Mulai	Tanggal Selesai	Prioritas	Deskripsi
Sprint 1	PB-01	Authentication System	2026-02-23	2026-03-15	P0 - Blocker	Implementasi login multi-role (GKM/GJM) dan session management
	PB-02	Dashboard GKM	2026-02-23	2026-03-15	P1 - High	monitoring RPS dan materi perkuliahan
	PB-03	Data Master Academic	2026-02-23	2026-03-15	P0 - Blocker	Halaman kelola Penugasan Dosen, Mata Kuliah, dan Periode Aktif
	PB-04	Monitoring RPS	2026-02-23	2026-03-15	P0 - Blocker	Integrasi data unggahan RPS dari API CIS Institut Teknologi Del
	PB-05	Reminder Upload RPS	2026-02-23	2026-03-15	P2 - Medium	Penyusunan pesan reminder personal menggunakan generator AI Agent
	PB-06	History Reminder RPS	2026-02-23	2026-03-15	P3 - Low	Pencarian dan log riwayat status pengiriman pengingat RPS
Sprint 2	PB-07	Monitoring Materi Perkuliahan	2026-03-23	2026-04-05	P0 - Blocker	Sinkronisasi upload materi (teori/praktikum) dari API CIS
	PB-08	Reminder Upload Materi	2026-03-23	2026-04-05	P2 - Medium	Generate teks pengingat materi otomatis berbasis AI
	PB-09	History Reminder Materi	2026-03-23	2026-04-05	P3 - Low	Halaman riwayat log dan statistik pengiriman pengingat materi
	PB-10	Pengelolaan Kuesioner	2026-03-23	2026-04-05	P0 - Blocker	Excel kuesioner dan analisis sentimen/poin oleh AI
	PB-11	Laporan Bulanan GKM	2026-03-23	2026-04-05	P1 - High	Otomatisasi draf laporan bulanan GKM berbasis data AI Agent
Sprint 3	PB-12	Laporan Kuesioner	2026-04-06	2026-04-19	P1 - High	Penyusunan infografis dan teks evaluasi kuesioner mahasiswa oleh AI
	PB-13	Reminder Agent	2026-04-06	2026-04-19	P2 - Medium	Sistem penjadwalan otomatis pengiriman pesan
	PB-14	Kirim Pesan GKM	2026-04-06	2026-04-19	P3 - Low	pengirima laporan oleh GKM ke pihak yang berkepentingan
	PB-15	Dashboard GJM	2026-04-06	2026-04-19	P1 - High	Dashboard kompilasi data mutu seluruh program studi fakultas vokasi
Sprint 4	PB-16	Laporan Triwulan	2026-04-20	2026-05-10	P1 - High	Ekstraksi data laporan GKM dengan OCR dan kompilasi laporan triwulan
	PB-17	Laporan Semester	2026-04-20	2026-05-10	P1 - High	implementasi pembuatan laporan Semester
	PB-18	Laporan VMTS	2026-04-20	2026-05-10	P1 - High	implementasi pembuatan laporan VMTS
	PB-19	Generate PPT Otomatis	2026-04-20	2026-05-10	P1 - High	Ekspor konten draf laporan langsung menjadi slide presentasi .pptx
	PB-20	Kirim Pesan GJM	2026-04-20	2026-05-10	P3 - Low	Pengiriman laporan kepada pihak terkait

WORKFLOW AGENT-BASED REPORT ASSITANT



WBS METRICS CALCULATOR & HIERARCHICAL ROLL-UP

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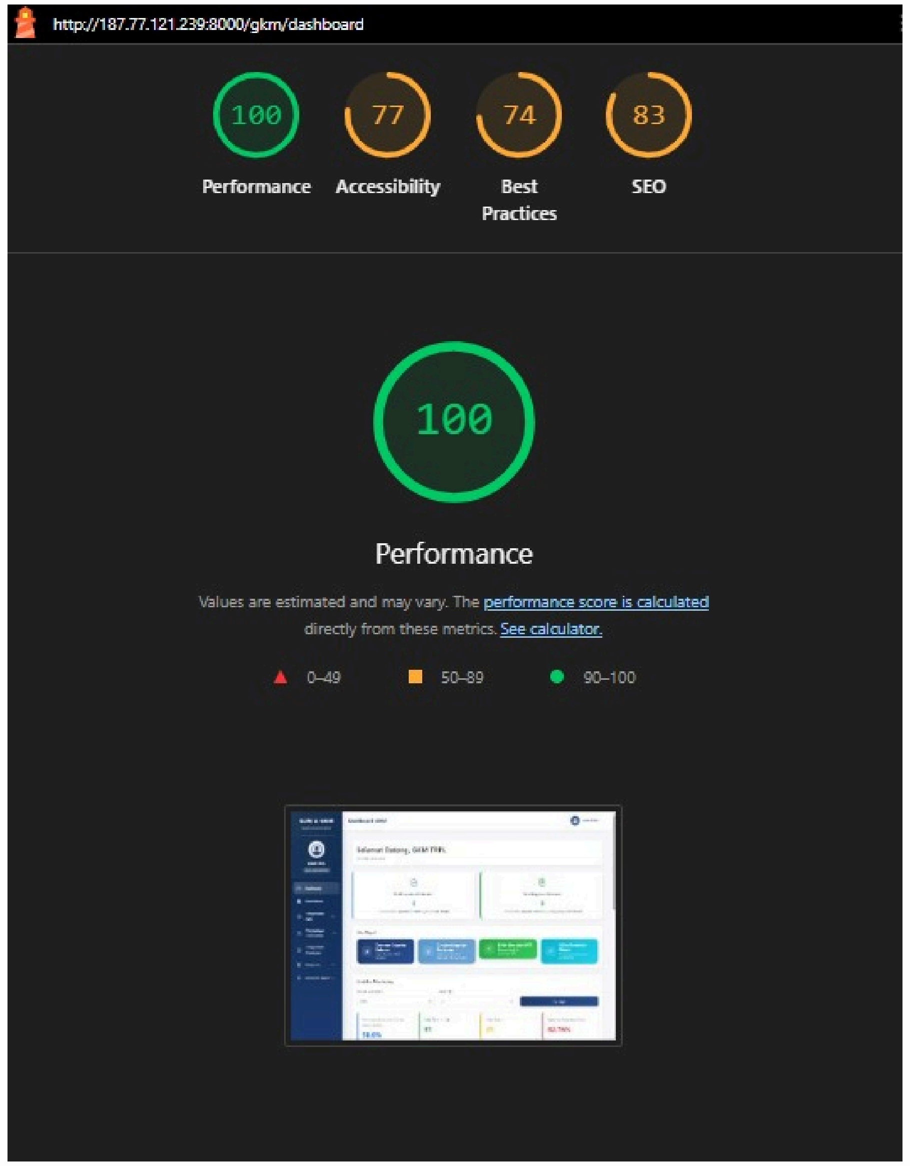
PBI AUTOMATIC CALCULATOR

PBI ID	Total Tasks	Total Substaks	Done Tasks	Progress %
PB-01	1	17	1	100.0%
PB-02	1	5	1	100.0%
PB-03	1	15	1	100.0%
PB-04	1	10	1	100.0%
PB-05	1	10	1	100.0%
PB-06	1	5	1	100.0%
PB-07	1	5	1	100.0%
PB-08	1	5	1	100.0%
PB-09	1	5	1	100.0%
PB-10	1	10	1	100.0%
PB-11	1	5	1	100.0%
PB-12	1	5	1	100.0%
PB-13	1	5	1	100.0%
PB-14	1	5	1	100.0%
PB-15	1	5	1	100.0%
PB-16	1	5	1	100.0%
PB-17	1	5	1	100.0%
PB-18	1	5	1	100.0%
PB-19	1	4	1	100.0%
PB-20	1	4	1	100.0%

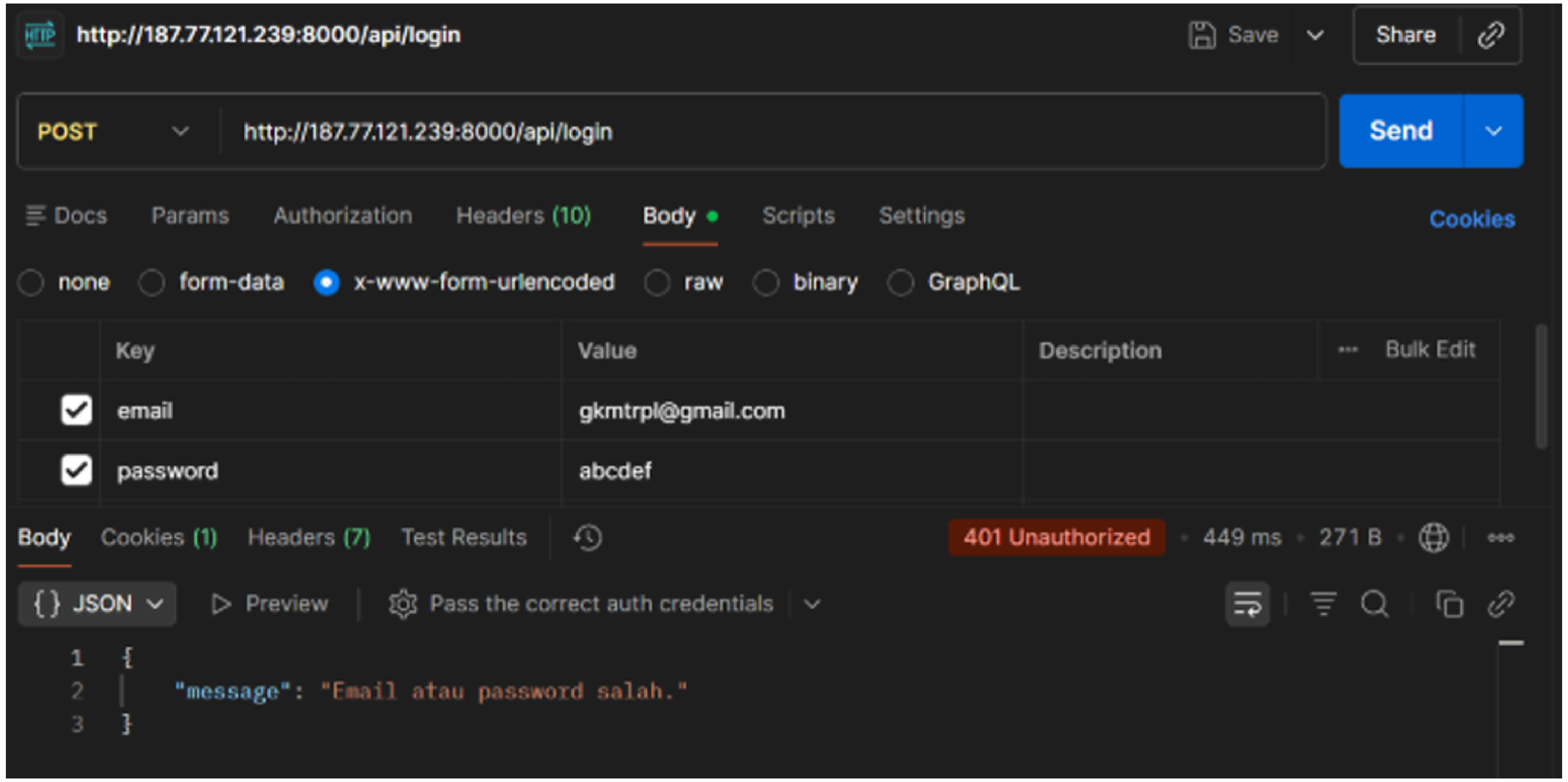
PROJECT COMPLETION OVERVIEW

In this project, we have developed a web-based agent system to automate GKM and GJM administration.
The system includes monitoring features, automatic reminders, questionnaire management, report generation, AI-based analysis, and automatic PPT creation.

TESTING RESULTS



All test cases passed after re-run



Postman testing confirmed that all API endpoints function correctly, returning valid responses and status codes.

MODEL EVALUATION

RAGAS Score Chart:

NO	PERTANYAAN / SKENARIO	KATEGORI	FAITH.	HALLU.	PREC.	RECALL	F1
1	Buat laporan triwulan terbaru	Laporan Triwulan	74.0%	15.0%	88.0%	91.0%	80.0%
2	perbaiki bagian pendahuluan	Laporan Bulanan	76.0%	10.0%	80.0%	87.0%	83.0%
3	Buatkan laporan VMTS semester ini	Laporan VMTS	73.0%	12.0%	78.0%	79.0%	73.0%
4	bagian tindak lanjut perbaiki	Laporan Bulanan	73.0%	17.0%	70.0%	78.0%	74.0%
5	Buatkan laporan kuesioner terbaru	Laporan Kuesioner	80.0%	11.0%	79.0%	84.0%	81.0%
6	perbaiki bagian pelaksanan dan ubah jadi 2026	Laporan Semester	79.0%	14.0%	75.0%	76.0%	70.0%
7	Buatkan laporan artefak bulanan	Laporan Bulanan	76.0%	13.0%	82.0%	89.0%	85.0%
8	Ringkas visi, misi, dan tujuan Program Studi TRPL	Laporan VMTS	73.0%	13.0%	87.0%	72.0%	89.0%
9	Buatkan laporan semester genap terbaru	Laporan Semester	81.0%	13.0%	75.0%	83.0%	79.0%
10	pada bagian rekomendasi masih belum bagus coba perbaiki lagi	Laporan Semester	70.0%	16.0%	77.0%	77.0%	72.0%
		Rata-rata	76.0%	13.0%	79.0%	82.0%	79.0%
		RAGAS Score	79.2%				

The system performed well, but showed room for improvement in the accuracy of generated answers (Faithfulness: 76%) Good

PROTOTYPE EVALUATION

SUS Score Bar Chart:

Responden	Total Kontribusi	Skor SUS	Kategori
R-01	31	77,50	Good (B)
R-02	20	67,50	Okay (C)
R-03	28	70,00	Good (B)
Rata-Rata		71,67	Good (B)

Average Score: 71.67 Category: Good (B) This indicates the system is considered usable.

UEQ Radar Chart atau Bar Chart:

Skala UEQ	Mean	Interpretasi
Attractiveness	2,06	Excellent
Perspicuity	2,08	Excellent
Efficiency	1,25	Excellent
Dependability	2,25	Excellent
Stimulation	2,33	Excellent
Novelty	2,58	Excellent

All six dimensions (Attractiveness, Perspicuity, Efficiency, Dependability, Stimulation, Novelty) scored in the Excellent category.

CONCLUSION

The project successfully build a fully functional system that automates the administrative processes of GKM and GJM. The system is considered a valuable tool for the digital transformation of academic quality assurance at the Faculty Vocational Studies Software Engineering Major in Institut Teknologi Del. Future Improvements: Further optimization can be done on the RAG system's response accuracy (Faithfulness) and some technical usability aspects.

DEMO

THANK YOU